Predicting Customer Booking Completion

British Airways Data Science Job Simulation - Task 2

APPROACH

- 50,000 customer booking records
- Feature engineering created 5 additional variables
- 80/20 stratified train-test split
- Route and booking origin target encoded
- Sales channel and trip type one-hot encoded
- Random Forest classifier (200 trees)
- 5-fold stratified cross-validation
- Class imbalance (15% positive) addressed via 0.20 probability threshold, prioritising recall

5-fold cross-validation

Mean ROC-AUC: 78.15% | Recall: 66.53%

Test performance remained close to CV performance.

MODEL PERFORMANCE

79.02%
ROC-AUC
5-fold Cross-Validated

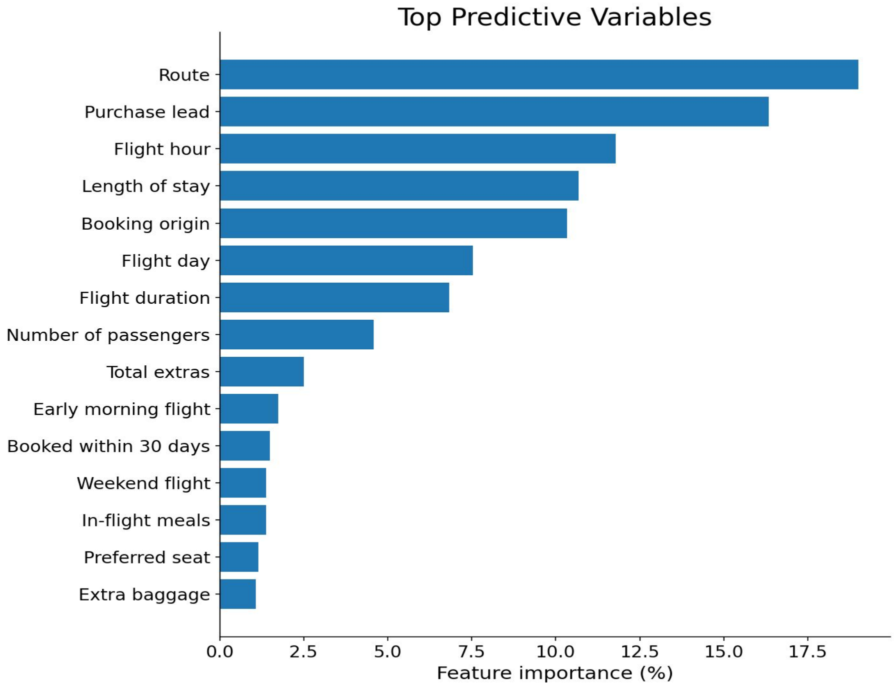
68.18%
Recall
of actual buyers correctly flagged

32.58%
Precision
of flagged customers actually book

15.00%
Baseline rate

Model flags 68% of actual bookers while targeting just 31% of all customers - useful for prioritizing marketing spend.

KEY PREDICTIVE VARIABLES



Feature importance (%) from the Random Forest model output.

Business Recommendations

1. **Target customers at the right time:** Customers booking within 30 days convert at 16.45% vs 14.18% otherwise. Retarget customers past the 30-day mark with timely campaigns.
2. **Use ancillary interest as an engagement signal:** Booking rates increased from 10.67% (no extras) to 18.60% (all three extras). Prioritise high-interest customers for targeted offers.
3. **Personalise campaigns by market and route:** Route and booking origin are top-5 predictive features. Build market/route-specific campaigns instead of one global message.

Note: Feature importance indicates predictive usefulness, not causation.